

INTRODUCTION

“Expressing One Life through Another”
Can living visual feedback of numerical data reshape the boundary between the self and others?

TEMOCHIMORI is a palm sized kokedama (“moss ball”) whose appearance reflects its wearer’s live heart rate.

Physiological biofeedback usually presents bodily change as numbers or graphs, leaving unclear whether a living display changes the experience. We present TEMOCHIMORI—literally “*holding a forest in your hand*” in Japanese—a living interface that visualizes your inner state, giving you an opportunity to feel connected with nature while inviting self-awareness.

PROTOTYPE

HEART RATE TO COLOR MAPPING

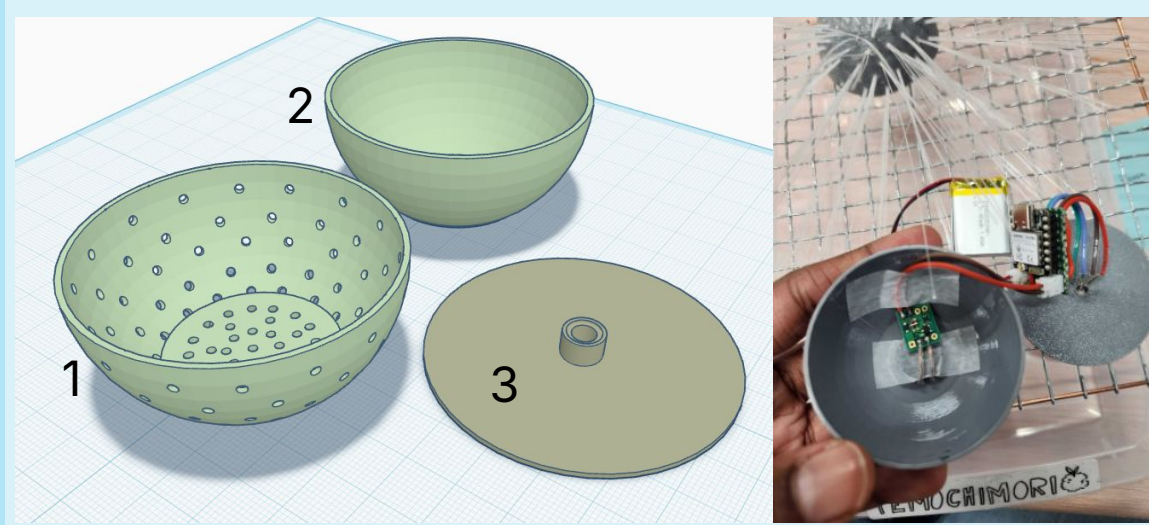
Blue
< 80 bpm

Green
80 - 90 bpm

Red
> 90 bpm

INNER SHELL

1. Top half with 112 openings for optical fibers
2. Bottom half to house electrical components
3. Middle divider to protect the electrical components and house the LED



WATERPROOFING

- Three liquid waterproofing coats on inner shell
- 24-hour immersion test → no water ingress into electronics compartment



ASSEMBLY

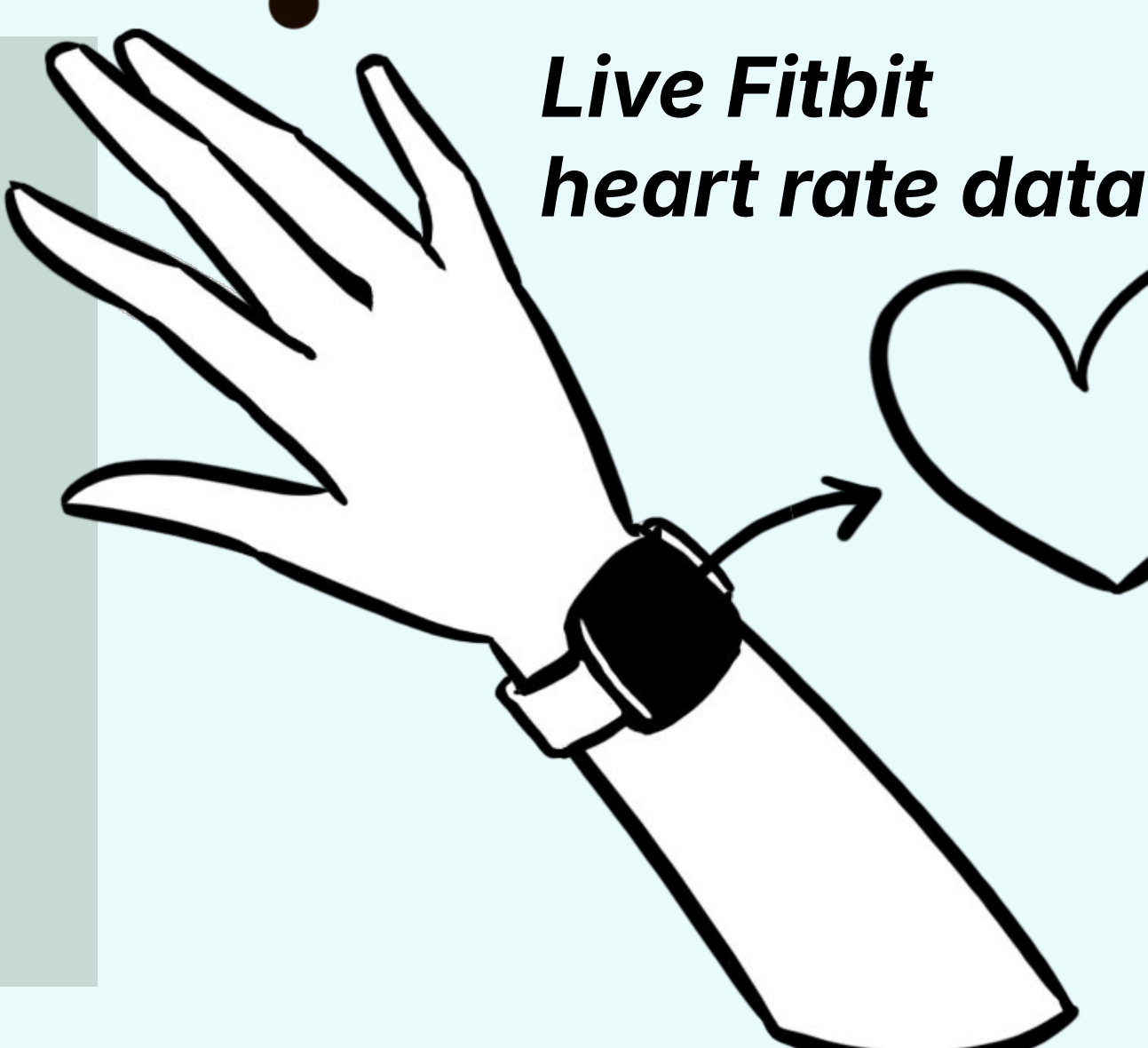
Soil covering, moss covering, fiber wire cutting



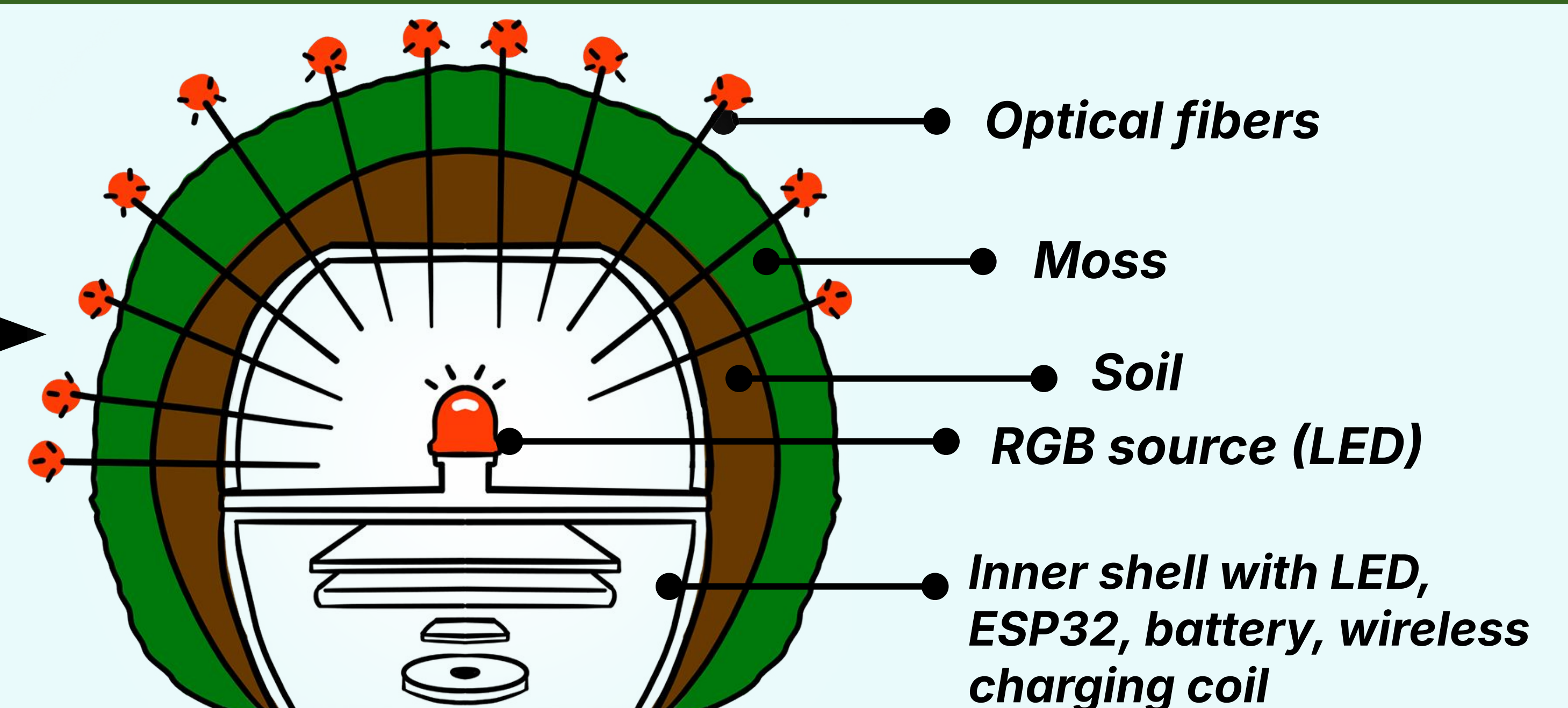
PROCESS

TEMOCHIMORI's **input** is the wearer's **live heart rate** derived from their Fitbit data.

With the ESP32 WiFi microcontroller, this data is translated to RGB values and visualized via optical fibers.



ESP32
WiFi microcontroller



RESULTS

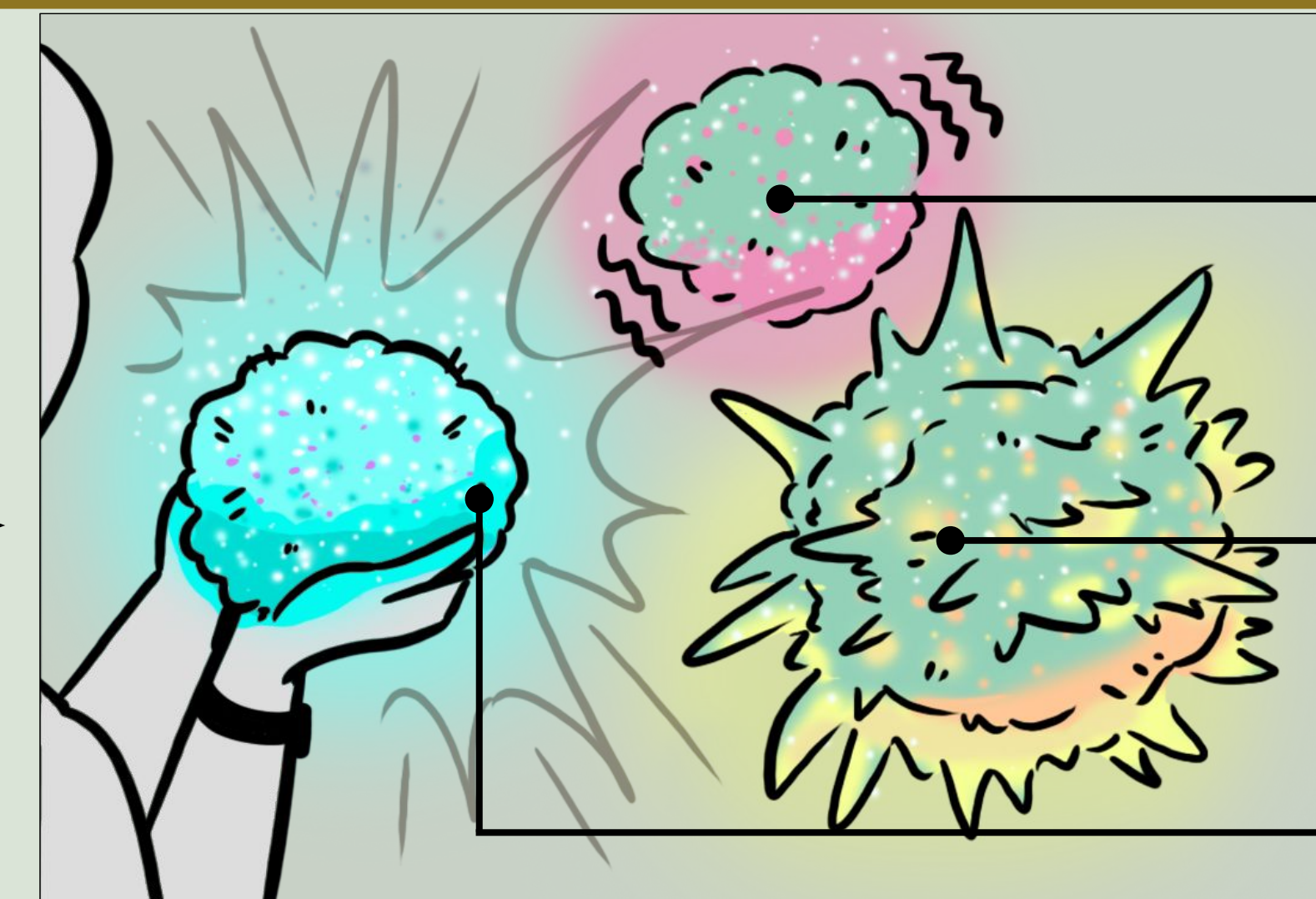
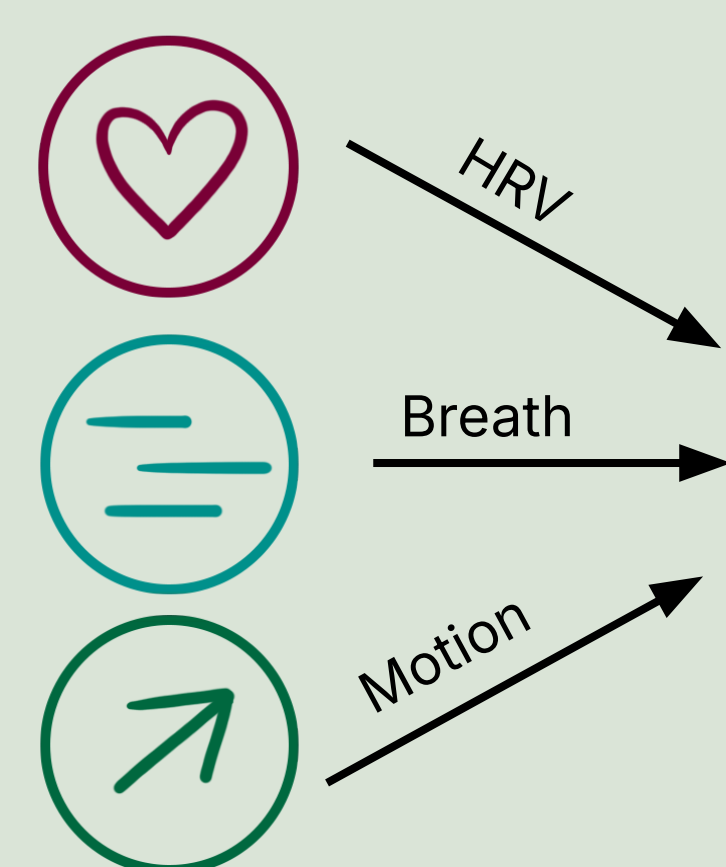
TEMOCHIMORI is a living artifact whose perceptible expression is continuously modulated by another organism's physiological signal

- Real-time reflection of Fitbit-derived heart rate confirmed
- The current prototype changes appearance, not moss physiology
- The hypothesized user-side loop is **perception → reflection / regulation → care**

FUTURE VISION

- Multi-signal input: HRV, respiration, movement
- Output: color, vibration, texture, expansion, contraction
- Self-awareness and emotion regulation through gentle living expression
- Social interaction mediated through TEMOCHIMORI

MULTI-SIGNAL INPUT



OUTPUT

VIBRATION

SHAPE

COLOR

Toward living interfaces that help people care for themselves by caring for another life

DISCUSSION

Our current contribution is a living kokedama (“moss ball”) that reflects a user’s Fitbit-derived heart rate. It is a real time design probe for how living visual feedback may reshape self/other boundaries and care practices. No user study has yet been conducted. Limitations include simplified three-color output, one-way coupling, a single immersion test, and a time consuming building process. Longitudinal evaluation should record moss hydration, visible condition, survival, enclosure integrity, and changes in users' perceptions, emotional responses, and care practices over time.

COLLABORATIONS

The Well-being Engineering Lab focuses on:

- Human-Nature Interaction
- Physiological Computing
- Biofeedback
- Living Well-being Interfaces

Check out our website for more information →



<https://www.welbe.imi.i.u-tokyo.ac.jp/en/>